Retail Data Platform – Requirements Document

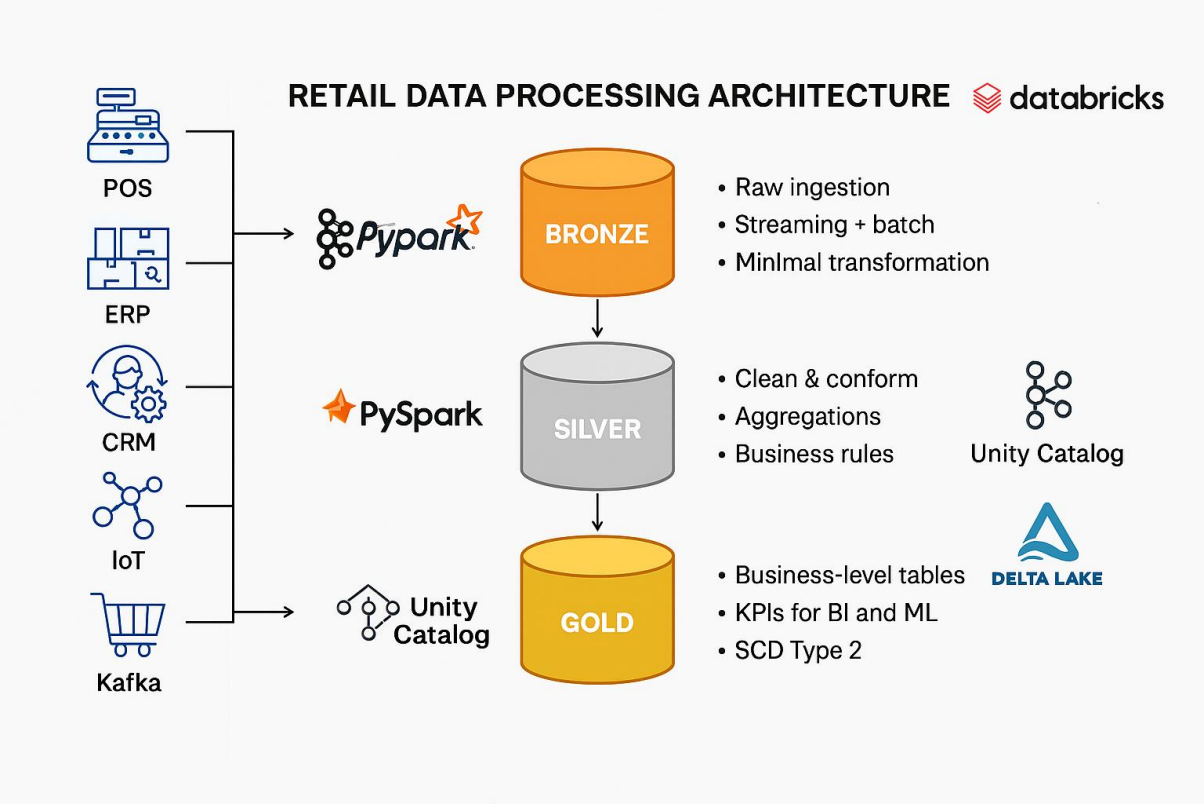
Platform: Databricks (Delta Lake, Unity Catalog, PySpark)
Industry: Retail (Omnichannel)
Architecture: Medallion (Bronze → Silver → Gold)

---

1. Purpose of the Document

This document defines the **business, functional, technical, data, and non-functional requirements** for building a **scalable, governed, and real-time retail analytics platform** using **Databricks and Delta Lake**, enabling:

- Historical and real-time retail analytics
- Customer, product, and inventory insights
- Regulatory compliance and data governance
- Support for BI, ML, and advanced analytics



---

2. Business Objectives

Primary Objectives

- Provide **single source of truth** for retail data

- Enable **real-time and batch analytics**
- Track **historical dimension changes (SCD Type-2)**
- Reduce data latency (< 5 minutes for streaming use cases)
- Support **self-service BI and ML**

Success Metrics

Metric	Target
Data availability SLA	≥ 99.9%
Streaming latency (Not Applicable for Offline data process)	< 2 minutes
Query performance	< 5 sec (Gold layer)
Data quality errors	< 0.5%

---

3. In-Scope Use Cases

Business Use Cases

1. Omnichannel Sales Analytics
2. Customer 360 View
3. Inventory Monitoring & Alerts
4. Product & Promotion Performance
5. Demand Forecasting (future phase)
6. Audit & Historical Reporting

Out of Scope (Phase-1)

- Third-party data monetization
- Unstructured social media data

---

4. Source Systems & Data Ingestion Requirements

Source Systems

Source	Type	Ingestion Mode

POS Systems	CSV / Kafka	Streaming
E-Commerce	API / CSV	Batch
CRM	CSV	Batch
ERP (Products)	CSV	Batch
IoT Inventory Sensors	Kafka	Streaming

---

5. Architecture Requirements

Medallion Architecture

BRONZE – Raw, Immutable

SILVER – Cleansed, Conformed

GOLD – Business Aggregates & Dimensions

Platform Constraints

- Databricks Runtime with Photon enabled
- Delta Lake as the only storage format
- Unity Catalog mandatory for governance

---

6. Data Layer Requirements

6.1 Bronze Layer Requirements

Purpose

Store **raw source data** with **full fidelity**.

Functional Requirements

- Append-only storage
- Support schema evolution
- Capture ingestion metadata

Mandatory Columns

Column	Description
ingestion_ts	Load timestamp
source_system	Origin
source_file	File name / topic

Sample Bronze Dataset – sales_raw.csv

CSV

order_id,order_date,store_id,product_id,customer_id,quantity,price

ORD1001,2026-04-20,STR01,PRD101,CUST01,2,500

ORD1002,2026-04-20,STR02,PRD102,CUST02,1,1200

Show more lines

6.2 Silver Layer Requirements

Purpose

Provide **clean, standardized, and conformed data**.

Functional Requirements

- Deduplication
- Data quality validation
- Business rule validation
- Reference data enrichment

Data Quality Rules

- quantity > 0
- price > 0
- customer_id NOT NULL

Silver Tables

- sales
- customers
- products

- inventory_events

Sample Silver Dataset – customers.csv

CSV

customer_id,name,email,city,loyalty_tier

CUST01,Rahul,rahul@gmail.com,Chennai,GOLD

CUST02,Anita,anita@gmail.com,Bangalore,SILVER

Show more lines

6.3 SCD Type-2 Dimension Requirements

Dimension: Customer

Business Need

Track historical changes for:

- City
- Loyalty Tier
- Contact details

Rules

1. One active record per business key
2. Changes create a new row
3. History must never be lost

Required Columns

Column	Description
customer_sk	Surrogate key
customer_id	Business key
effective_start_date	Start
effective_end_date	End

is_current	Current flag

6.4 Gold Layer Requirements

Purpose

Support **business reporting, dashboards, ML, and KPIs**.

Gold Tables

Table	Description
fact_sales	Transaction facts
dim_customer	SCD-2
dim_product	SCD-2
daily_store_sales	Aggregates

Sample Gold Dataset – fact_sales.csv

CSV

```
order_id,customer_id,store_id,product_id,sale_date,net_amount
ORD1001,CUST01,STR01,PRD101,2026-04-20,1000
ORD1002,CUST02,STR02,PRD102,2026-04-20,1200
```

Show more lines

---

7. Real-Time Processing Requirements

Streaming Scenarios

1. Real-time sales tracking
2. Inventory threshold alerts

SLA

- End-to-end latency < 2 minutes
- Exactly-once processing

Sample Streaming Dataset – inventory_events.csv

CSV

product_id,store_id,event_time,stock_level

PRD101,STR01,2026-04-20T10:30:00,5

PRD102,STR02,2026-04-20T10:32:00,45

Show more lines

---

8. Unity Catalog & Governance Requirements

Mandatory Controls

- Centralized metadata
- Column-level PII masking
- Role-based access
- Lineage tracking

Catalog Structure

```
retail_prod
 ├── bronze
 ├── silver
 └── gold
```

PII Fields

- email
- phone
- address

---

9. Performance & Optimization Requirements

Spark

- Enable Adaptive Query Execution
- Broadcast joins for small dimensions

Delta

- Partition by date
- Z-Order: customer_id, product_id

- OPTIMIZE weekly

---

10. Security & Compliance Requirements

- Encryption at rest (Delta)
- Encryption in transit
- GDPR/PII compliance
- Audit access logs

---

11. Non-Functional Requirements

Category	Requirement
Scalability	Handle 10x data growth
Availability	99.9%
Recovery	Time travel & rollback
Maintainability	Modular pipelines

---

12. Assumptions & Dependencies

Assumptions

- Source data adheres to agreed schema
- Business keys uniquely identify records

Dependencies

- CRM and POS data availability
- Network connectivity to Databricks

---

13. Acceptance Criteria

✅ Data loads successfully to Bronze
✅ Clean data available in Silver
✅ SCD Type-2 history maintained

✅ Gold tables BI-ready
✅ Unity Catalog enforcement

---

14. Deliverables

- Databricks notebooks (PySpark & SQL)
- Delta tables (Bronze/Silver/Gold)
- Sample datasets (CSV / Excel)
- Data model & ER diagram
- Power BI dashboards (optional)